



Birla Institute of Technology & Science, Pilani

Pilani Campus

AUGS/AGSR Division

CS F301 | Principles of Programming Languages

Module 01 — SYMPHONY Live City Twin

Lectures Industry briefing

Amit Dua

First Semester 2025–26



What this briefing is

- A live working digital twin for traffic, not a six-junction cartoon
- Operator selects a city pack: Bengaluru, London, New York, Hong Kong
- A fifth city is a pack file, not a fork of the command centre
- Patent envelope: Indian application 202611024014, claims 1–10



Outcomes

- Switch city and show the map, timezone, and agencies change
- Name the live adapters and what happens when a key is missing
- Trace one 30-second cycle from catalogue to advisory
- State Observe, Recommend, and Actuate without mixing them

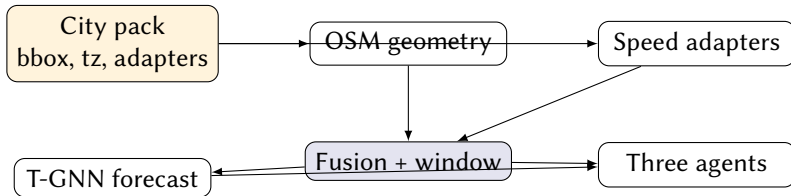


The problem the patent addresses

- Fixed-time signals ignore the next half hour of demand
- One model at every junction cannot meet a sub-second budget and a one-hour plan at once
- Cities will not buy a system that dies when one feed drops
- Partners outside Bengaluru will not accept Silk Board hard-coded



Claim 1 in one diagram



- Three language models, distinct in size and latency, assigned by timescale
- Forecasts are an input to junction, corridor, and network agents
- Signal controllers keep a local fail-safe (claim 7)



City pack: the contract

- id, tz, bbox, agencies, corridor, adapters
- Geometry: OpenStreetMap, cached for Pages, refresh in Recommend
- Speeds: whatever keys that city grants (Google, TomTom, TfL, NYC 511, data.gov.hk)
- Cameras: optional catalogue. HLS or WHEP in the browser. Never RTSP JSON



Four packs shipping today

- **Bengaluru.** ORR Silk Board to Marathahalli. BTP / GBA. Google + TomTom when keyed. Open-Meteo always
- **London.** A501 Marylebone to King's Cross. TfL Unified API
- **New York.** FDR Brooklyn Bridge to 96th Street. NYC Open Data
- **Hong Kong.** Island East Central to Quarry Bay. TD / data.gov.hk



Three operating modes

- **Observe.** Live APIs + OSM. Advisories only. Public Pages
- **Recommend.** Same plus logged phase and offset suggestions. Partner with keys
- **Actuate.** Physical or SUMO write-back. Licensed pilot only. Blocked in this build



30-second cycle (as filed)

- 0–5 s collect adapters in parallel
- 5–8 s fuse: freshness, accuracy, consistency; 3σ down-weight
- 8–12 s T-GNN 5 / 15 / 30 minute speeds
- 12–25 s network, corridor, junction decisions
- 25–28 s command generation (advisory here)
- 28–30 s predicted versus observed (claim 10)



Timescales and model scale

- Junction (73–78): smallest, quantized, ≤ 200 ms (claims 6, 7)
- Corridor (72): intermediate, minutes, offsets from predicted speed (claim 8)
- Network (71): largest, slowest, hours. Incidents, VIP, pattern shift (claim 9)
- If the centre drops past the outage interval, local stored logic runs



What “live” means in this demo

- Source health panel: green / amber / red, age in seconds
- Open-Meteo precipitation is a real HTTP call (Bengaluru 200 today)
- NYC Open Data traffic speeds: HTTP 200 on i4gi-tjb9
- Google / TomTom stay off Pages (no keys in the browser)
- Missing keys do not blank the map. The twin keeps running



Cameras, if a city grants them

- Start from a catalogue GET, never a guessed /stream/id
- RTSP over TCP for inference. HLS if 8554 is blocked. WHEP for preview
- Timing from PTS, never arrival time, never declared FPS
- Backoff 2 s to 30 s. Decode warnings at join are logs, not fatal
- Consume only. Cap four open captures



Command centre, this evening

- Open #/app. Select city. Clock follows that timezone
- Overview: health, 30 s cycle, six junction cards, agent feed
- Switch London: Marble Arch to King's Cross, not Silk Board
- Cameras tab: honest empty state when the pack has no catalogue
- Actuate option is labelled blocked



Evidence already on file

- Delay -65% , speed $+17\%$, queue -72% versus fixed-time (FER reply, $p < 0.001$, 15 trials)
- Spatial MAE 3.23 km/h, temporal MAE 6.01 km/h
- Quick-test harness is a separate table. Do not mix it with 352% throughput from the longer A/B design



What a partner can do next

- Observe on Pages today, no install
- Recommend: supply Google/TomTom or a city token. Keys stay on a backend
- Actuate: licence 202611024014, then a controller adapter
- Fifth city: one pack, OSM bbox, one corridor, optional catalogue



Where the bits live

- App: <https://amitduabits.github.io/symphony-digital-twin/#/app>
- Source: <https://github.com/amitduabits/symphony-digital-twin>
- Live-city book: `docs/LIVE_CITY_PROMPTBOOK.md`
- Applicant: Yushu Excellence Technologies Pvt. Ltd.
- Inventor: Amit Dua, CSIS, BITS Pilani